

C++2: Lesson 3 – Lab Project

1. Add **summary** at the top and **comments** where necessary.
- Recordings and interviews are an exception, but they should be included in your submission code.
2. **Cite** the source of your information.
3. Refrain from including anything **you don't fully understand**.

2 **100 points**

Modify **Question #3 in Lesson 2** to update the program as follows:

1. Add an overloaded constructor to initialize the **radius** directly when creating a **Circle** object.
2. Add overloaded functions for calculating the area and circumference, which accept a radius as a parameter. These functions should not modify the object's **radius** but instead return results based on the provided parameter.
3. Update the program to demonstrate both:
Calculations using the object's radius.
Calculations using an alternate radius (e.g., 10.0) through the overloaded functions.

The modified program should:

-Prompt the user for a circle's radius.

-Use the Circle class to calculate and display:

The area, diameter, and circumference using the object's radius.

The area and circumference for an alternate radius using the overloaded functions.

Sample
output

Radius? 5

Radius: 5.00

Area: 78.54 square units

Diameter: 10.00 units

Circumference: 31.42 units

Results using an alternate radius of 10.0:

Area: 314.16 square units

Circumference: 62.83 units

File(s) to
submit

C203_2.cpp

C203_2.png